Why Topology Still Matters When the Measurement Is Not Noise Immune

Noise, chain length, and the optimal VQE depth

Seminar notes

July 16, 2026

Abstract

The Kitaev-chain simulation is not noise immune in the NISQ sense: the state is prepared by ordinary noisy gates, and the edge-string diagnostic is a long, non-local measurement whose contrast is easily reduced by gate, readout, and decoherence errors. Topology still matters because it changes the *ideal* many-body physics: the logical information is stored non-locally in two separated Majorana edge modes, and the finite-size splitting between the two parity sectors falls exponentially with chain length. The practical rule is therefore not "make the circuit deeper." It is: use the *shallowest adequately expressive* ansatz depth. For the current $L = 4$ study this was $r = 2$; for larger L the minimum useful value of r may increase, but every extra repetition also adds $L - 1$ more noisy CNOTs.

1 What topology protects

For even L, the Jordan–Wigner parity operator used in the code is

$$\hat{P} = \prod_{j=0}^{L-1} Z_j. \tag{1}$$

The Kitaev-chain Hamiltonian is

$$H = -\frac{\mu}{2}\sum_j Z_j + \frac{t-\Delta}{2}\sum_j X_j X_{j+1} + \frac{t+\Delta}{2}\sum_j Y_j Y_{j+1}. \tag{2}$$

Every term contains either zero or two X/Y operators, so $[H, \hat{P}] = 0$. Parity conservation itself is therefore a $\mathbb{Z}_2$ symmetry of the Hamiltonian, not a separate miracle of topology.

Topology enters in a different way. In the topological phase, $|\mu| < 2t$, the chain supports Majorana modes at opposite ends. The two end modes form one non-local fermion. A local perturbation acting near one end cannot fully measure or split that non-local degree of freedom unless it effectively connects the two ends or creates a parity-changing event. In a finite chain the two edge modes overlap weakly, so the even/odd splitting is not exactly zero, but it is exponentially small:

$$E_0(L) \approx 2t(1-\lambda^2)\lambda^L, \qquad \lambda = \frac{|\mu|}{2t}. \tag{3}$$

This is the useful protection: topology suppresses the *intrinsic* Hamiltonian splitting as L grows, while the bulk remains gapped away from the critical points.

2 Why this is not noise immune

The phrase "topologically protected" is easy to overstate. It does not mean that any laboratory implementation or any measurement circuit is immune to all noise. In this project there are four separate reasons the measured signal is not noise immune.

1. The chain is finite. At finite L the Majorana wavefunctions overlap. The ideal splitting is small in the topological phase, but it is not exactly zero. Increasing L reduces this overlap, but it does not remove device noise from the circuit used to prepare and measure the state.

2. The VQE circuit is an ordinary noisy circuit. The topological Hamiltonian is simulated with standard qubits and standard gates. The ansatz used in the code is

```
EfficientSU2(L, su2_gates=['ry'], entanglement='linear', reps=r)
```

Each repetition adds a linear CNOT layer. Thus

$$N_\theta = L(r+1), \qquad N_{\mathrm{CNOT}} = r(L-1). \tag{4}$$

If each CNOT carries an error channel, the topological nature of the target state does not stop those channels from acting. The noisy simulation in `block4.py` correctly applies the channel after every transpiled `cx` gate.

3. Some noise violates the protecting symmetry. Phase damping is diagonal in the computational basis, so it commutes with $\hat{P}$ and preserves parity. Amplitude damping and depolarizing noise do not: they contain operations that change the local Z_j eigenvalue and therefore can flip the global parity. In the repository convention, $c_j^\dagger c_j = (I + Z_j)/2$, so computational $|0\rangle$ is the occupied single-site state; the important point is not whether a channel is described as adding or removing a fermion, but that it changes fermion-number parity.

4. The measured topological diagnostic is non-local. The edge string

$$O_{\mathrm{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1} \tag{5}$$

is the correct finite-size Majorana edge diagnostic, but it is a long Pauli string. Long strings are fragile under measurement noise. Symmetric readout error with bit-flip probability ϵ attenuates a weight-L string roughly as

$$(1-2\epsilon)^L. \tag{6}$$

Dephasing can leave $\langle \hat{P} \rangle$ unchanged while still destroying the endpoint coherence that the edge string needs. This is why Block 4 can show both facts at once: parity may remain fixed under parity-preserving noise, while the topological string signal decays.

3 Why topology still matters

The model is still important because it provides a passive physical layer of protection that ordinary local qubits do not have. A local qubit stores information in one local degree of freedom, so a local disturbance can directly read or damage it. A Majorana pair stores information non-locally, so local perturbations have exponentially small access to the encoded splitting when the bulk gap is open and the chain is long.

That is not a replacement for error correction, calibration, or careful measurement. It is a different starting point: the ideal Hamiltonian already suppresses one important class of errors. The NISQ problem is that our present simulation does not use native protected Majorana hardware. It synthesizes the state with ordinary CNOTs, then measures a long string. The physics has topological structure; the circuit implementation still has ordinary gate noise.

This is also why the model is useful as a benchmark. It lets us separate three questions that are often mixed together:

1. Does the ideal Hamiltonian have topological edge modes?
2. Can a variational circuit prepare the corresponding ground state?
3. Does a noisy device preserve enough contrast to measure the non-local diagnostic?

The answer can be yes, yes, and only partially. That is a meaningful result, not a contradiction.

4 How increasing chain length changes the optimal reps

Let

$$S_{\text{meas}}(L, r) = |\langle O_{\text{edge}} \rangle|_{\text{meas}} \tag{7}$$

be the measured edge-string magnitude after preparing the state with `reps`$= r$. A useful approximation is

$$S_{\text{meas}}(L, r) \approx S_{\text{ideal}}(L, r) \exp[-\eta_{\text{cx}}\, r(L-1)] \exp[-\eta_{\text{ro}}\, L], \tag{8}$$

where $S_{\text{ideal}}(L, r)$ is the noiseless expressibility of the ansatz, $\eta_{\text{cx}} = -\log(1 - p_{\text{cx}}) \approx p_{\text{cx}}$ is the per-CNOT noise cost, and $\eta_{\text{ro}} \approx -\log(1 - 2\epsilon)$ is the readout cost per measured qubit. The constants depend on the channel, but the structure is robust: expressibility improves with r, while noise worsens with the total number of CNOTs, $r(L-1)$.

Equation (8) gives the practical rule. Increasing the reps from r to $r+1$ is worthwhile only if

$$\log \frac{S_{\text{ideal}}(L, r+1)}{S_{\text{ideal}}(L, r)} > \eta_{\text{cx}}(L-1). \tag{9}$$

The left side is the expressibility gain from one more layer. The right side is the extra noise cost of the additional $L-1$ CNOTs. As L grows, the cost of one extra repetition grows linearly. Therefore deeper circuits become harder to justify at larger L.

What happened at $L = 4$

For the current $L = 4$ depth scan, $r = 1$ is under-expressive: the noiseless edge string is essentially zero. At $r = 2$, the ansatz first reaches the correct topological state, and deeper circuits do not significantly improve the ideal signal. Once the ideal signal has saturated, Eq. (9) fails: the numerator is near one, while the extra CNOT cost is positive. Thus the noisy optimum is

$$r_{\text{opt}} = 2 \qquad (L = 4,\ \mu = 0,\ \text{current ansatz}). \tag{10}$$

This is why the Week 9 result is described as the *shallowest adequately expressive depth.*

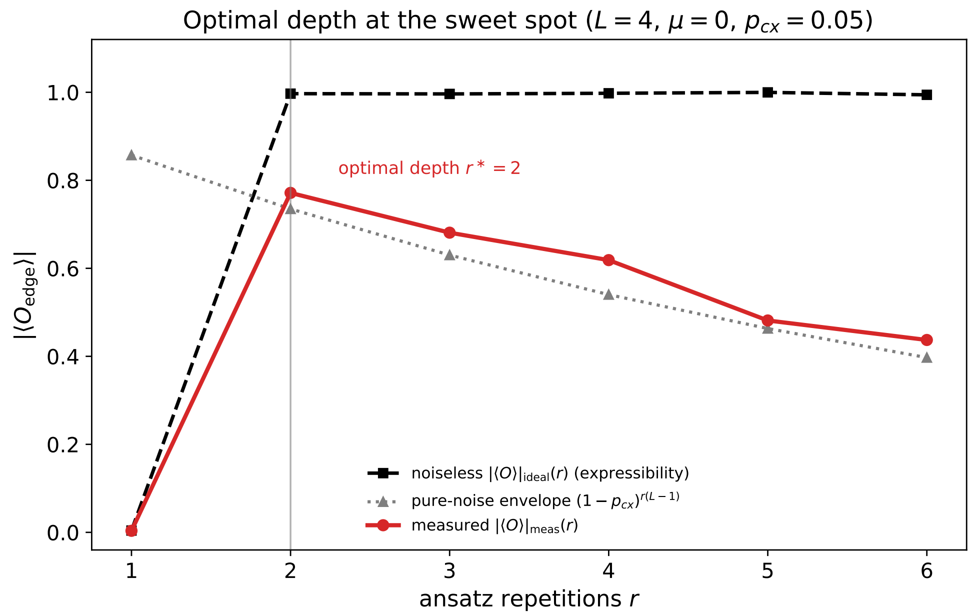

Figure 1: For $L = 4$, the noiseless edge string has an expressibility threshold: $r = 1$ fails, $r = 2$ succeeds, and deeper circuits mainly accumulate CNOT noise.

Should r increase when L increases?

Usually, the minimum expressive depth should be expected to stay the same or increase with L, not decrease. Larger chains contain more bonds, more variational parameters, more parity constraints, and a longer non-local string. Near the critical points $|\mu| = 2t$, the correlation length grows, so the ground state is harder to represent accurately. A larger r may therefore be needed to clear the same energy, fidelity, parity, and edge-string tolerances.

But there is no universal formula such as $r = L/2$ for this hardware-efficient ansatz. The required depth depends on the ansatz, the phase point μ, the target tolerance, and the optimizer. Deep in the gapped topological phase, a small number of repetitions may remain enough for the small system sizes we can simulate. Close to criticality, or if high global fidelity is required across a longer chain, the needed r can grow.

The correct definition is empirical and validation-based:

$$r_{\text{expr}}(L, \mu) = \min \left\{ r : \begin{array}{l} |E_r - E_{\text{ED}}| \leq \delta_E, \\ |\langle \hat{P} \rangle_r - 1| \leq \delta_P, \\ 1 - F_r \leq \delta_F, \\ |S_{\text{ideal}}(L, r) - S_{\text{ED}}(L)| \leq \delta_S \end{array} \right\}. \tag{11}$$

Then, when the ideal signal has threshold-like behavior, the noisy optimum is approximately

$$r_{\text{opt}}(L, \mu) \approx r_{\text{expr}}(L, \mu). \tag{12}$$

In words: choose the first depth that passes the noiseless validation, and stop there unless one more layer gives a visible ideal-signal gain larger than the extra noise penalty in Eq. (9).

5 The chain-length trade-off

Increasing L helps and hurts at the same time.

Effect of increasing L	Scaling tendency	Consequence
Majorana overlap	$\sim (\lvert\mu\rvert/2t)^L$	better intrinsic topological protection
Readout/string weight	$\sim (1-2\epsilon)^L$	weaker measured string contrast
CNOT count at fixed r	$r(L-1)$	more accumulated gate noise
CNOT count if $r_{\text{expr}} \sim L$	$\sim L^2$	severe noisy-depth penalty

If $r_{\text{expr}}(L)$ stays approximately constant, the circuit-noise envelope still falls roughly as

$$\exp[-\eta_{\text{cx}}\, r_{\text{expr}} L]. \tag{13}$$

If the expressive depth itself grows linearly with L, then

$$N_{\text{CNOT}} = r_{\text{expr}}(L)(L-1) = O(L^2), \tag{14}$$

and the measured signal can collapse much faster:

$$S_{\text{meas}} \sim \exp[-\text{const}\cdot L^2]. \tag{15}$$

This is the central NISQ tension. Topology improves the ideal finite-size physics with L, but the circuit used to demonstrate that physics becomes longer and noisier with L.

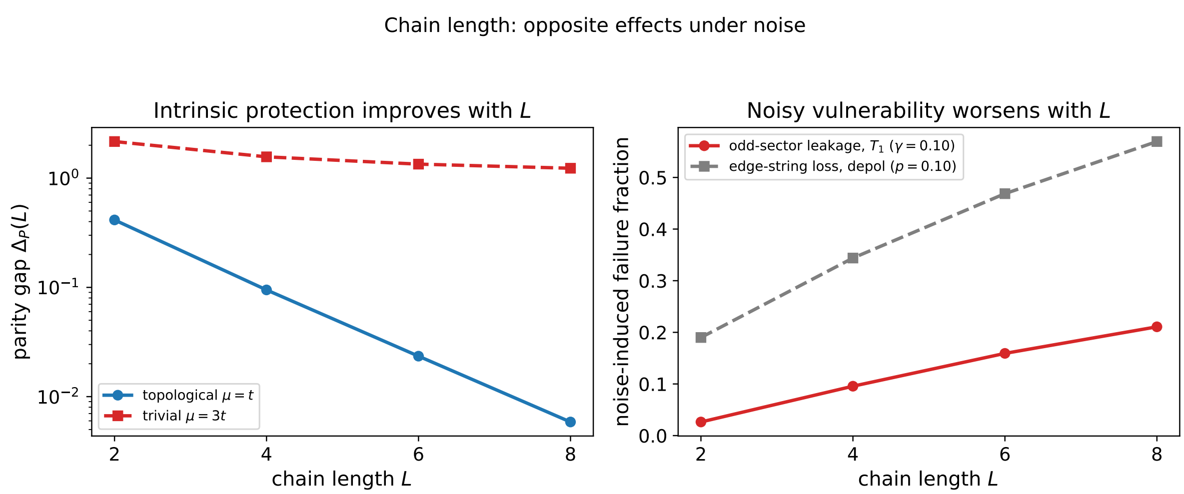

Figure 2: Longer chains reduce the intrinsic topological splitting but increase exposure to noise in parity-changing and string-measurement channels.

6 Practical answer for the project

For $L = 4$, the data justify $r = 2$ as the optimum under CNOT noise because it is the first depth that prepares the state. For larger L, do not assume $r = 2$ will always be enough, but also do not increase r automatically. Run the same depth scan at each target L. In the noisy study, choose

$$r_{\text{opt}} = \underset{r\ \text{passing validation}}{\arg\max}\ S_{\text{meas}}(L, r). \tag{16}$$

With the threshold behavior observed here, this usually means the shallowest depth that passes the ED/VQE validation and still gives the largest noisy edge-string contrast.

If the depth scan says that r_{expr} grows quickly with L, then the generic hardware-efficient ansatz is becoming the bottleneck. The right response is not simply to keep adding layers. Better options are a more Hamiltonian-inspired ansatz, parity-preserving gates, adiabatic/matchgate-style state preparation, or restricting the NISQ demonstration to the largest L for which the edge-string contrast remains measurable.

7 Summary

- The system is not noise immune because finite chains have overlap, simulated Majorana states are prepared by ordinary noisy gates, parity-violating channels can poison the protected sector, and the edge string is a long non-local measurement.
- Topology still matters because it stores information non-locally and suppresses the ideal even/odd splitting exponentially with L.
- The useful signal is approximately $S_{\mathrm{ideal}}(L, r)\exp[-\eta_{\mathrm{cx}} r(L-1)]$ times readout loss.
- Increasing r is useful only while the ideal expressibility gain beats the extra CNOT-noise cost. At $L = 4$, $r = 2$ is enough; for larger L, r_{expr} may increase, but every added repetition costs $L - 1$ more CNOTs.
- The correct rule is not "larger L means larger r"; it is "larger L requires a new depth scan, and the optimum is the shallowest validated depth."

References

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[3] J. Alicea, *New directions in the pursuit of Majorana fermions in solid state systems*, arXiv:1202.1293.